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High-Performance State Synchronization in Edge-Native Autonomous Networks

Decentralized Consensus via Dynamic Epoch Partitioning in the Nexa Architecture

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Abstract

Modern edge-native autonomous applications require sub-hundred-millisecond state synchronization across heterogeneous, bandwidth-constrained node topologies. Traditional Byzantine Fault Tolerant (BFT) protocols and Raft-based consensus mechanisms suffer from scalability bottlenecks when deployed over highly dynamic, lossy edge links. In this paper, we present the **Meridian Protocol**, a novel state synchronization engine built for the Nexa Autonomous Network framework. Meridian introduces *Dynamic Epoch Partitioning* (DEP), a lightweight consensus mechanism that decouples local state commit phases from global deterministic ordering. By organizing edge nodes into localized Synapse clusters and leveraging Merkle State Delta (MSD) trees, Meridian achieves strict serializability with minimal message overhead. We evaluate Meridian on a 500-node testbed deployed across Apex Cloud and Nimbus Edge infrastructure. Our empirical results demonstrate a $4.2\times$ increase in transactional throughput (up to 48,500 ops/sec) and a 68% reduction in P99 synchronization latency compared to state-of-the-art Raft variants under 15% packet loss conditions.

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CCS Concepts: Computer systems organization → Distributed architectures; Reliable execution; Software and its engineering → Distributed systems infrastructure.

Keywords: Edge Computing, Distributed Consensus, Fault Tolerance, State Synchronization, Merkle Trees, Nexa Architecture.

1. INTRODUCTION

The proliferation of autonomous physical systems—ranging from automated robotic fleets deployed by Meridian Logistics to real-time industrial telemetry networks operated by Nexa Infrastructure—has shifted the locus of data processing from centralized cloud data centers toward heterogeneous edge nodes [1]. These edge-native environments are characterized by intermittent network connectivity, constrained compute capabilities, and strict sub-hundred-millisecond operational latency budgets.

Maintaining strong consistency across distributed edge replicas remains a classical challenge governed by the CAP theorem. Traditional consensus algorithms such as Raft and PBFT guarantee linearizability by requiring a majority quorum for every state mutation [2]. However, when network partitions or high latency jitter occur over regional backhaul links, quorum acquisition stalls, leading to catastrophic queueing delays and service degradation.

To address these limitations, we introduce the **Meridian Protocol**, a high-throughput state synchronization architecture engineered specifically for the Nexa Autonomous Network ecosystem. Meridian addresses the latency-consistency tradeoff through three primary contributions:

- **Dynamic Epoch Partitioning (DEP):** A hierarchical state delegation model that isolates transient state